

Homework 03

Helena Parsons

Part 1: Setting up tasks

Reading in packages

```
library(tidyverse)
library(here)
library(janitor)
library(readxl)
```

Creating data object

```
# creating new object for temp-kelp data
kelp <-
  # reading csv
  read_csv(
    # directing function to data file
    here("data", "temp-kelp.csv"))

# creating new object for personal data
job_search <-
  # reading csv
  read_csv(
    # directing function to data file
    here("data", "personal-data.csv"))
```

Part 2: Problems

Problem 1: Giant kelp fronds

*Climate change threatens all ecosystems, including giant kelp (*Macrocystis pyrifera*) forests.*

You are interested in the potential relationship between ocean temperature (measured in °C) and giant kelp frond elongation rate (also known as growth rate, measured in cm day⁻¹).

You conduct a field study in which you track giant kelp individuals growing in different temperatures (on average) and measure their elongation rates. Admittedly, this is not a perfect study, but you do the best with what you have!

1a. An appropriate test

In 1-3 sentences, name the appropriate test(s) to determine the strength of the relationship between temperature and giant kelp frond elongation rate (hint: there are two). Describe the differences between the two tests.

Be specific in your response to demonstrate your understanding of the variables in this question.

Because I solely want to understand the strength of the relationship between temperature and giant kelp frond elongation rate, I want to use a correlation test.

Correlation gives you a value for how strongly two variables are related, which is why it should be used for this analysis.

Multiple linear regression can be used when you want to understand the strength of a relationship, but is more commonly used to estimate the response variable value when an explanatory variable is at a certain value.

1b. Create a visualization

Create a visualization that would be appropriate for showing the relationship between temperature and giant kelp frond elongation rate.

In addition to using the correct geometries, be sure to:

- *relabel the x- and y-axes and include units*
- *use different colors from the ggplot() defaults*
- *use a different theme from the ggplot() default*